CLI Usage

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1、CLI Introduction

CLI (Command-Line Interface) is a user interface that interacts with computer systems or programs through keyboard text commands. It was the mainstream interaction method before the popularity of Graphical User Interfaces (GUIs) and is still widely used in system administration, automation scripts, development tools, and other fields.

2、Enable Maximum Board Performance

2.1、Enable MAX Power Mode

Enabling MAX Power Mode on Jetson will ensure all CPU and GPU cores are enabled:

```
sudo nvpmodel -m 0
```

2.2、Enable Jetson Clocks

Enabling Jetson Clocks will ensure all CPU and GPU cores run at maximum frequency:

```
sudo jetson_clocks
```

3、Using CLI for Prediction

The following CLI commands are all run in the desktop terminal.

3.1、CLI Usage Syntax

```
yolo TASK MODE ARGS

Where   TASK (optional) is one of [detect, segment, classify, pose, obb]
        MODE (required) is one of [train, val, predict, export, track,
benchmark]
        ARGS (optional) are any number of custom 'arg=value' pairs like
'imgsz=320' that override defaults.
```

3.2、Image Prediction

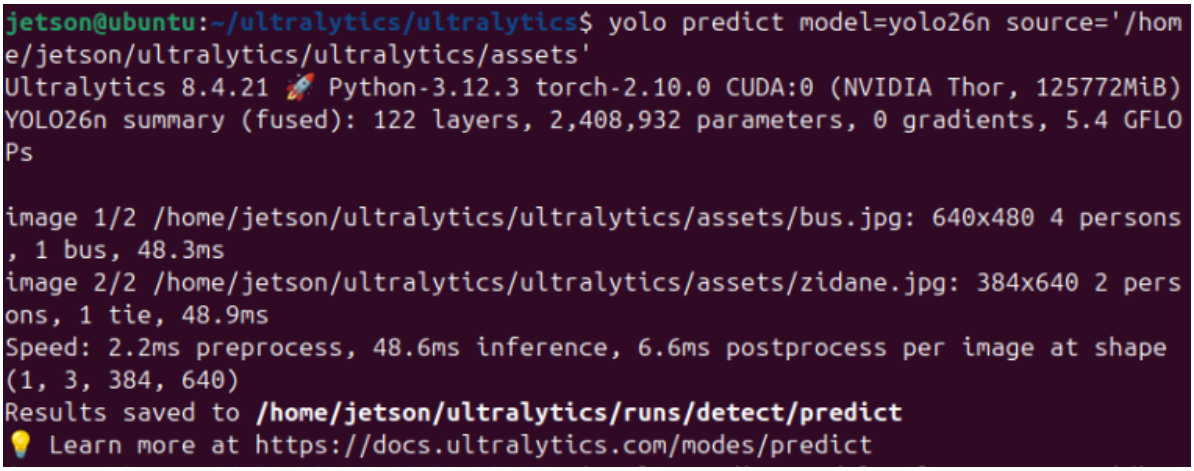
Use yolo26n.pt to predict images included in the ultralytics project: If the system cannot find the corresponding model file in the directory where the command is run, it will automatically download (if download fails, you can copy the model in manually)

Enter project folder:

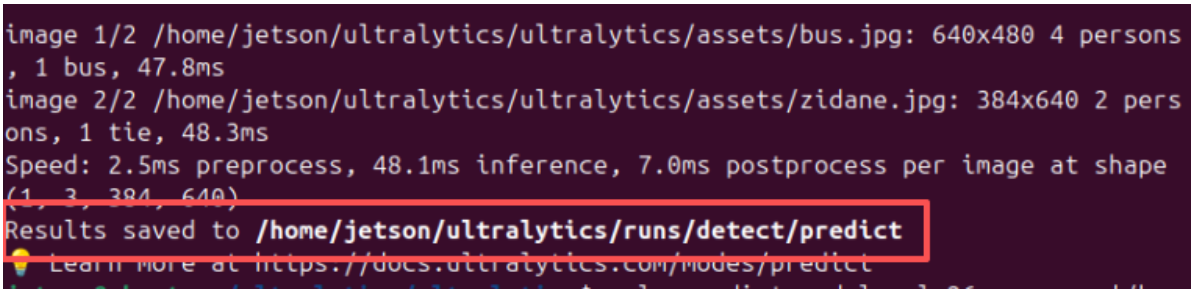
```
cd /home/jetson/ultralytics/ultralytics/
```

Use yolo26n.pt to detect images in the target folder and output results:

```
yolo predict model=yolo26n source='/home/jetson/ultralytics/ultralytics/assets'
```



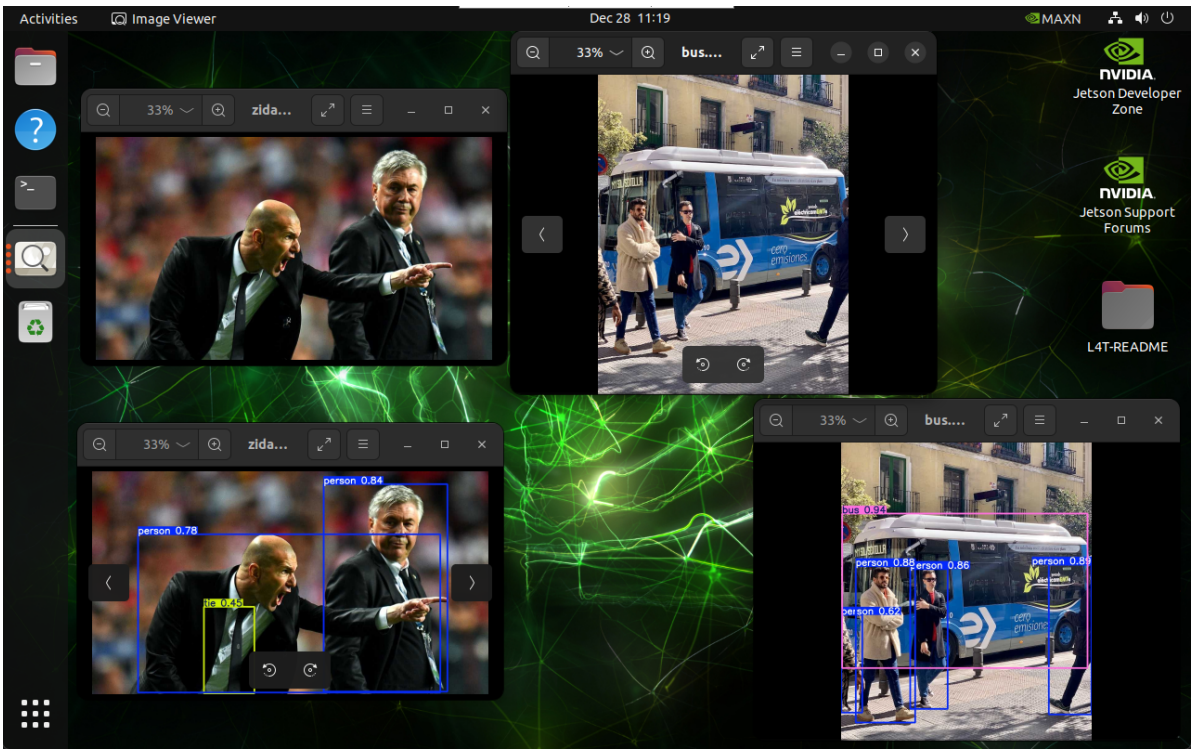
Note: Each time it runs, the predict in the output folder automatically increments by 1. Record the path printed by the program.



Effect Preview

yolo recognition output location: /home/jetson/ultralytics/runs/detect

Note: The actual path should be based on the path printed above. /home/jetson/ultralytics/runs/detect/predict



3.3、Video Prediction

Use yolo26n.pt to predict videos in the ultralytics project (not videos that come with ultralytics, if the program cannot find the video, you need to manually place the video file in the /home/jetson/ultralytics/ultralytics/videos path): If the system cannot find the corresponding model file in the directory where the command is run, it will automatically download (if download fails, you can copy the model in manually)

Enter project folder:

```
cd /home/jetson/ultralytics/ultralytics
```

Use yolo26n.pt to detect videos in the target folder and output results:

```
yolo predict model=yolo26n source='/home/jetson/ultralytics/ultralytics/videos'
```

```
jetson@ubuntu:~/ultralytics/ultralytics$ yolo predict model=yolo26n source='/hom
e/jetson/ultralytics/ultralytics/videos'
Ultralytics 8.4.21 🚀 Python-3.12.3 torch-2.10.0 CUDA:0 (NVIDIA Thor, 125772MiB)
YOLO26n summary (fused): 122 layers, 2,408,932 parameters, 0 gradients, 5.4 GFLO
Ps

video 1/5 (frame 1/180) /home/jetson/ultralytics/ultralytics/videos/car.mp4: 384
x640 12 cars, 1 bus, 48.5ms
video 1/5 (frame 2/180) /home/jetson/ultralytics/ultralytics/videos/car.mp4: 384
x640 15 cars, 1 bus, 7.1ms
video 1/5 (frame 3/180) /home/jetson/ultralytics/ultralytics/videos/car.mp4: 384
x640 13 cars, 1 bus, 7.6ms
video 1/5 (frame 4/180) /home/jetson/ultralytics/ultralytics/videos/car.mp4: 384
x640 14 cars, 1 bus, 7.3ms
video 1/5 (frame 5/180) /home/jetson/ultralytics/ultralytics/videos/car.mp4: 384
x640 15 cars, 1 bus, 8.1ms
```

Each time it runs, the predict in this folder automatically increments by 1. Record the path printed by the program.

```
video 5/5 (frame 414/414) /home/jetson/ultralytics/ultralytics/videos/street.mp4
: 480x640 3 cars, 3 trucks, 7.4ms
Speed: 1.6ms preprocess, 7.5ms inference, 0.4ms postprocess per image at shape (
1, 3, 480, 640)
Results saved to /home/jetson/ultralytics/runs/detect/predict2
💡 Learn more at https://docs.ultralytics.com/modes/predict
jetson@ubuntu:~/ultralytics/ultralytics$
```

Effect Preview

yolo recognition output video location: /home/jetson/ultralytics/runs/detect

You can use ffmpeg to view videos. Use the following command to install:

```
sudo apt install ffmpeg -y
```

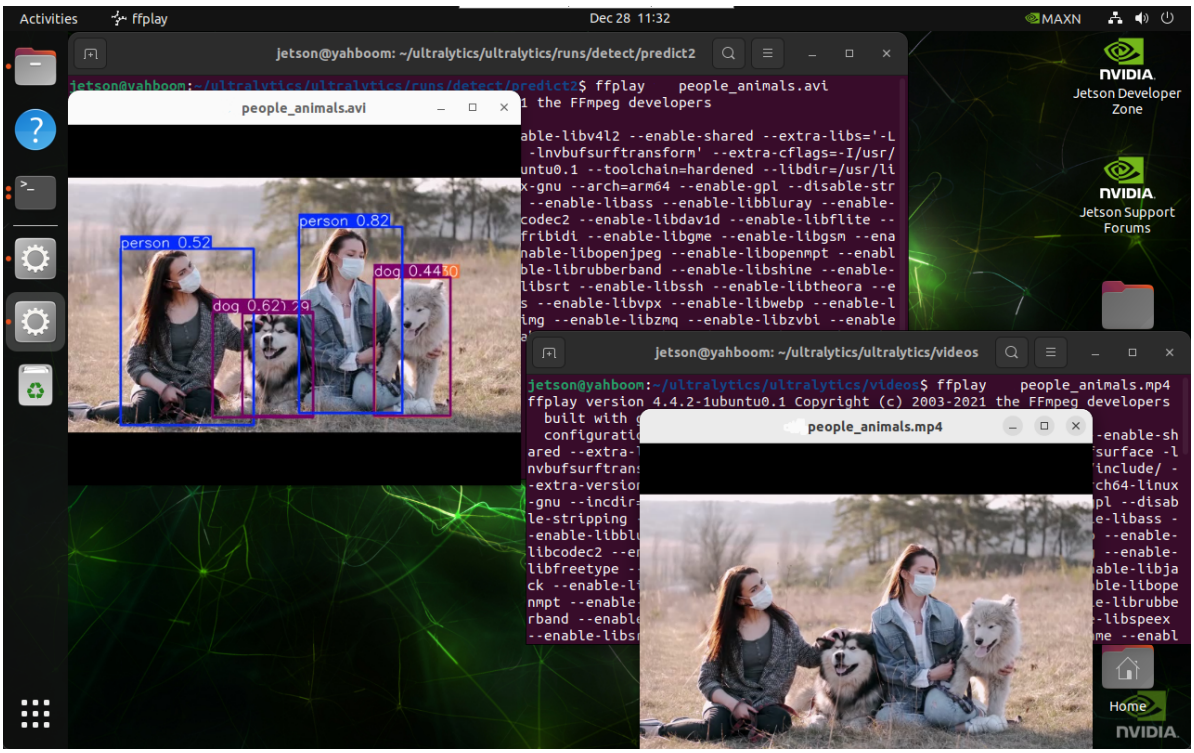
Enter video input path:

Note: Since predict automatically increments by 1 each time the program runs, use the actual output as the standard. predict2 needs to be changed to the actual folder name.

```
cd /home/jetson/ultralytics/ultralytics/runs/detect/predict2
```

Play video:

```
ffplay people_animals.avi
```



3.4、Real-time Prediction

Please first connect the USB camera to the USB port of the motherboard.

Use yolo26n.pt to predict USB camera footage: If the system cannot find the corresponding model file in the directory where the command is run, it will automatically download (if download fails, you can copy the model in manually)

Enter project folder:

```
cd /home/jetson/ultralytics/ultralytics
```

Use yolo26n.pt to detect camera footage and output results. You can choose one of the following functions to run.

After the program runs, it will open the camera footage and detect it in real-time, displaying the output results on the screen.

```
# Object detection
yolo predict model=yolo26n.pt source=0 save=False show

# Instance segmentation:
yolo predict model=yolo26n-seg.pt source=0 save=False show

# Image classification:
yolo predict model=yolo26n-cls.pt source=0 save=False show

# Pose estimation:
yolo predict model=yolo26n-pose.pt source=0 save=False show

# Oriented object detection:
yolo predict model=yolo26n-obb.pt source=0 save=False show
```

If this is the first time running the program, the model needs to be loaded, so the loading time will be relatively long. Please wait patiently.

Click the terminal and press the "Ctrl + C" shortcut to terminate the program!

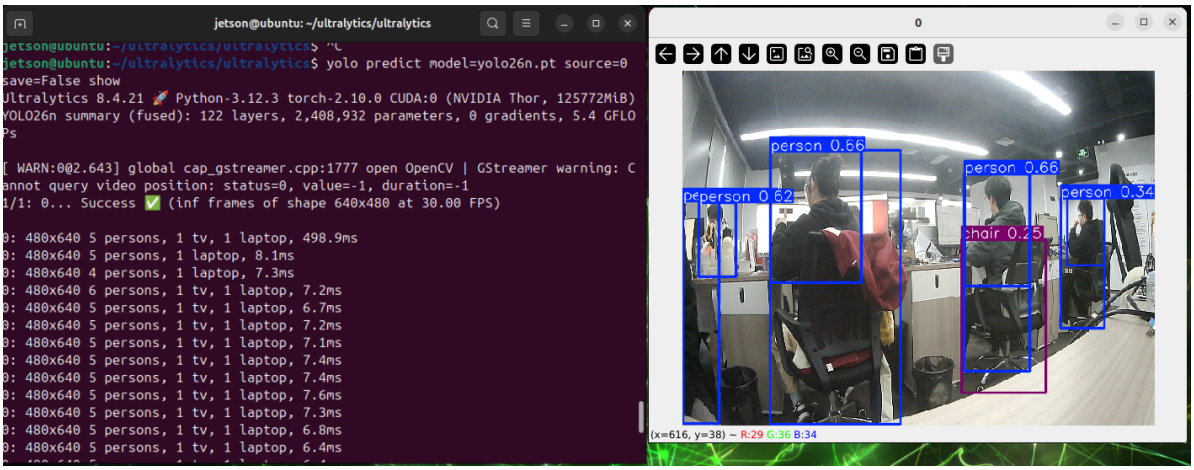
> **Parameter Description**

`model`: Specify YOLO model

`source`: Specify recognition source: multiple cameras can switch numbers

`save=False`: Disable saving results

`show`: Real-time display



References

https://github.com/ultralytics/ultralytics

https://docs.ultralytics.com/usage/cli/